

Trabalho Prático

Projeto e Implementação de Redes IPv4

Endereçamento IPv4, Roteamento Estático e Configuração via CLI no Cisco Packet Tracer

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RA (Registro Acadêmico)	2562790 (dois últimos dígitos: 90)
Modalidade	Individual
Esquema de endereçamento	200.200.90.x (N = 90)
Disciplina	Redes de Computadores – Prof. Eden Dosciatti

Cenário. A empresa **TechBuild Engenharia** está expandindo suas operações e necessita interligar filiais por meio de roteadores. Cada filial possui sua própria LAN e deve comunicar-se com todas as demais. O projeto foi desenvolvido em quatro fases (topologias), utilizando roteadores Cisco modelo **2811** e switches modelo **2960**, com endereçamento IPv4 e rotas estáticas configuradas via linha de comando (CLI).

Pato Branco – PR · 2026

1. Planejamento de Endereçamento das Sub-redes

O esquema de endereçamento utiliza o bloco 200.200.90.x, onde o valor **N = 90** corresponde aos dois últimos dígitos do RA. A tabela abaixo apresenta as seis sub-redes do projeto, com seus respectivos endereços de rede, máscaras e faixas de hosts válidos.

Sub-Rede	Endereço da Sub-rede	Máscara	Faixa de hosts válidos
LAN-ARPANET	200.200.90.0	255.255.255.192 (/26)	.1 – .62
LAN-DARPA	200.200.90.64	255.255.255.224 (/27)	.65 – .94
LAN-WWW	200.200.90.96	255.255.255.224 (/27)	.97 – .126
WAN-IMP-NET	200.200.90.128	255.255.255.252 (/30)	.129 – .130
WAN-MILNET	200.200.90.132	255.255.255.252 (/30)	.133 – .134
WAN-ARPA	200.200.90.136	255.255.255.252 (/30)	.137 – .138

Regras de atribuição de endereços

- **PC-Baran, PC-Cerf e PC-TimBerners:** primeiro endereço de host válido da respectiva LAN.
- **PC-Kleinrock, PC-Kahn e PC-Mockapetris:** segundo endereço de host válido.
- **Switches (SW-IMP, SW-Ethernet, SW-RadiaPerlman):** penúltimo endereço de host válido (na SVI VLAN 1).
- **Interface do roteador conectada à LAN:** último endereço de host válido (atua como *gateway* da LAN).
- **Enlaces WAN (/30):** primeiro e segundo endereços de host válidos, um em cada extremidade do enlace.

2. Tarefa 1 – Endereçamento das Interfaces

Tabelas de endereçamento IPv4 de todos os dispositivos, para cada uma das quatro topologias.

Topologia 1

Dispositivo	Interface	IPv4	Máscara de sub-rede	IPv4 Gateway
PC-Baran	NIC	200.200.90.1	255.255.255.192 (/26)	200.200.90.62
PC-Kleinrock	NIC	200.200.90.2	255.255.255.192 (/26)	200.200.90.62
PC-Cerf	NIC	200.200.90.65	255.255.255.224 (/27)	200.200.90.94
PC-Kahn	NIC	200.200.90.66	255.255.255.224 (/27)	200.200.90.94
SW-IMP	SVI	200.200.90.61	255.255.255.192 (/26)	200.200.90.62
SW-Ethernet	SVI	200.200.90.93	255.255.255.224 (/27)	200.200.90.94
RT-Taylor	Fa0/0	200.200.90.62	255.255.255.192 (/26)	—
RT-Taylor	Fa0/1	200.200.90.129	255.255.255.252 (/30)	—
RT-Postel	Fa0/0	200.200.90.130	255.255.255.252 (/30)	—
RT-Postel	Fa0/1	200.200.90.94	255.255.255.224 (/27)	—

Topologia 2

Dispositivo	Interface	IPv4	Máscara de sub-rede	IPv4 Gateway
PC-Baran	NIC	200.200.90.1	255.255.255.192 (/26)	200.200.90.62
PC-Kleinrock	NIC	200.200.90.2	255.255.255.192 (/26)	200.200.90.62
PC-Cerf	NIC	200.200.90.65	255.255.255.224 (/27)	200.200.90.94
PC-Kahn	NIC	200.200.90.66	255.255.255.224 (/27)	200.200.90.94
SW-IMP	SVI	200.200.90.61	255.255.255.192 (/26)	200.200.90.62
SW-Ethernet	SVI	200.200.90.93	255.255.255.224 (/27)	200.200.90.94
RT-Taylor	Fa0/0	200.200.90.62	255.255.255.192 (/26)	—
RT-Taylor	Fa0/1	200.200.90.129	255.255.255.252 (/30)	—
RT-Postel	Fa0/0	200.200.90.134	255.255.255.252 (/30)	—
RT-Postel	Fa0/1	200.200.90.94	255.255.255.224 (/27)	—
RT-Babbage	Fa0/0	200.200.90.130	255.255.255.252 (/30)	—
RT-Babbage	Fa0/1	200.200.90.133	255.255.255.252 (/30)	—

Topologia 3

Dispositivo	Interface	IPv4	Máscara de sub-rede	IPv4 Gateway
PC-Baran	NIC	200.200.90.1	255.255.255.192 (/26)	200.200.90.62
PC-Kleinrock	NIC	200.200.90.2	255.255.255.192 (/26)	200.200.90.62
PC-Cerf	NIC	200.200.90.65	255.255.255.224 (/27)	200.200.90.94
PC-Kahn	NIC	200.200.90.66	255.255.255.224 (/27)	200.200.90.94

Dispositivo	Interface	IPv4	Máscara de sub-rede	IPv4 Gateway
PC-TimBerners	NIC	200.200.90.97	255.255.255.224 (/27)	200.200.90.126
PC-Mockapetris	NIC	200.200.90.98	255.255.255.224 (/27)	200.200.90.126
SW-IMP	SVI	200.200.90.61	255.255.255.192 (/26)	200.200.90.62
SW-Ethernet	SVI	200.200.90.93	255.255.255.224 (/27)	200.200.90.94
SW-RadiaPerlman	SVI	200.200.90.125	255.255.255.224 (/27)	200.200.90.126
RT-Taylor	Fa0/0	200.200.90.62	255.255.255.192 (/26)	—
RT-Taylor	Fa0/1	200.200.90.129	255.255.255.252 (/30)	—
RT-Postel	Fa0/0	200.200.90.130	255.255.255.252 (/30)	—
RT-Postel	Fa0/1	200.200.90.94	255.255.255.224 (/27)	—
RT-Postel	Fa1/0	200.200.90.126	255.255.255.224 (/27)	—

Topologia 4

Dispositivo	Interface	IPv4	Máscara de sub-rede	IPv4 Gateway
PC-Baran	NIC	200.200.90.1	255.255.255.192 (/26)	200.200.90.62
PC-Kleinrock	NIC	200.200.90.2	255.255.255.192 (/26)	200.200.90.62
PC-Cerf	NIC	200.200.90.65	255.255.255.224 (/27)	200.200.90.94
PC-Kahn	NIC	200.200.90.66	255.255.255.224 (/27)	200.200.90.94
PC-TimBerners	NIC	200.200.90.97	255.255.255.224 (/27)	200.200.90.126
PC-Mockapetris	NIC	200.200.90.98	255.255.255.224 (/27)	200.200.90.126
SW-IMP	SVI	200.200.90.61	255.255.255.192 (/26)	200.200.90.62
SW-Ethernet	SVI	200.200.90.93	255.255.255.224 (/27)	200.200.90.94
SW-RadiaPerlman	SVI	200.200.90.125	255.255.255.224 (/27)	200.200.90.126
RT-Taylor	Fa0/0	200.200.90.62	255.255.255.192 (/26)	—
RT-Taylor	Fa1/0	200.200.90.137	255.255.255.252 (/30)	—
RT-Taylor	Fa0/1	200.200.90.129	255.255.255.252 (/30)	—
RT-Postel	Fa0/0	200.200.90.130	255.255.255.252 (/30)	—
RT-Postel	Fa1/0	200.200.90.133	255.255.255.252 (/30)	—
RT-Postel	Fa0/1	200.200.90.94	255.255.255.224 (/27)	—
RT-Babbage	Fa0/0	200.200.90.134	255.255.255.252 (/30)	—
RT-Babbage	Fa0/1	200.200.90.138	255.255.255.252 (/30)	—
RT-Babbage	Fa1/0	200.200.90.126	255.255.255.224 (/27)	—

Verificação das interfaces da Topologia 4 (show ip interface brief)

```
RT-Babbage-PS>enable
RT-Babbage-PS#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    200.200.90.134  YES manual up          up
FastEthernet0/1    200.200.90.138  YES manual up          up
FastEthernet1/0    200.200.90.126  YES manual up          up
Vlan1              unassigned      YES unset  administratively down down
RT-Babbage-PS#
```

Topologia 4 – topo4-briefBabbage.png

```
RT-Postel-PS>enable
RT-Postel-PS#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    200.200.90.130  YES manual up          up
FastEthernet0/1    200.200.90.94   YES manual up          up
FastEthernet1/0    200.200.90.133  YES manual up          up
Vlan1              unassigned      YES unset  administratively down down
RT-Postel-PS#
```

Topologia 4 – topo4-briefPostel.png

```
RT-Taylor-PS>
RT-Taylor-PS>enable
RT-Taylor-PS#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    200.200.90.62   YES manual up          up
FastEthernet0/1    200.200.90.129  YES manual up          up
FastEthernet1/0    200.200.90.137  YES manual up          up
Vlan1              unassigned      YES unset  administratively down down
RT-Taylor-PS#
```

Topologia 4 – topo4-briefTaylor.png

3. Tarefa 2 – Tabelas de Roteamento

Tipo: **C** = rede diretamente conectada · **S** = rota estática.

Topologia 1

RT-Taylor – Tabela de Roteamento				
Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.0	255.255.255.192 (/26)	—	Fa0/0
C	200.200.90.128	255.255.255.252 (/30)	—	Fa0/1
S	200.200.90.64	255.255.255.224 (/27)	200.200.90.130	Fa0/1

RT-Postel – Tabela de Roteamento				
Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.64	255.255.255.224 (/27)	—	Fa0/1
C	200.200.90.128	255.255.255.252 (/30)	—	Fa0/0
S	200.200.90.0	255.255.255.192 (/26)	200.200.90.129	Fa0/0

Comprovação no Packet Tracer (show ip route)

```
RT-Postel-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 5 subnets, 4 masks
S       200.200.90.0/26 [1/0] via 200.200.90.129
C       200.200.90.64/27 is directly connected, FastEthernet0/1
L       200.200.90.94/32 is directly connected, FastEthernet0/1
C       200.200.90.128/30 is directly connected, FastEthernet0/0
L       200.200.90.130/32 is directly connected, FastEthernet0/0

RT-Postel-PS>
```

Topologia 1 — topo1-rotaPostel.png

```
RT-Taylor-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 5 subnets, 4 masks
C       200.200.90.0/26 is directly connected, FastEthernet0/0
L       200.200.90.62/32 is directly connected, FastEthernet0/0
S       200.200.90.64/27 [1/0] via 200.200.90.130
C       200.200.90.128/30 is directly connected, FastEthernet0/1
L       200.200.90.129/32 is directly connected, FastEthernet0/1

RT-Taylor-PS>
```

Topologia 1 — topo1-rotaTaylor.png

Topologia 2

RT-Taylor – Tabela de Roteamento

Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.0	255.255.255.192 (/26)	—	Fa0/0
C	200.200.90.128	255.255.255.252 (/30)	—	Fa0/1
S	200.200.90.132	255.255.255.252 (/30)	200.200.90.130	Fa0/1
S	200.200.90.64	255.255.255.224 (/27)	200.200.90.130	Fa0/1

RT-Babbage – Tabela de Roteamento

Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.128	255.255.255.252 (/30)	—	Fa0/0
C	200.200.90.132	255.255.255.252 (/30)	—	Fa0/1
S	200.200.90.0	255.255.255.192 (/26)	200.200.90.129	Fa0/0
S	200.200.90.64	255.255.255.224 (/27)	200.200.90.134	Fa0/1

RT-Postel – Tabela de Roteamento

Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.64	255.255.255.224 (/27)	—	Fa0/1
C	200.200.90.132	255.255.255.252 (/30)	—	Fa0/0
S	200.200.90.128	255.255.255.252 (/30)	200.200.90.133	Fa0/0
S	200.200.90.0	255.255.255.192 (/26)	200.200.90.133	Fa0/0

Comprovação no Packet Tracer (show ip route)

```
RT-Babbage-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 6 subnets, 4 masks
S       200.200.90.0/26 [1/0] via 200.200.90.129
S       200.200.90.64/27 [1/0] via 200.200.90.134
C       200.200.90.128/30 is directly connected, FastEthernet0/0
L       200.200.90.130/32 is directly connected, FastEthernet0/0
C       200.200.90.132/30 is directly connected, FastEthernet0/1
L       200.200.90.133/32 is directly connected, FastEthernet0/1

RT-Babbage-PS>
```

Topologia 2 — topo2-rotaBabbage.png

```
RT-Postel-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 6 subnets, 4 masks
S       200.200.90.0/26 [1/0] via 200.200.90.133
C       200.200.90.64/27 is directly connected, FastEthernet0/1
L       200.200.90.94/32 is directly connected, FastEthernet0/1
S       200.200.90.128/30 [1/0] via 200.200.90.133
C       200.200.90.132/30 is directly connected, FastEthernet0/0
L       200.200.90.134/32 is directly connected, FastEthernet0/0

RT-Postel-PS>
```

Topologia 2 — topo2-rotaPostel.png

```
RT-Taylor-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 6 subnets, 4 masks
C       200.200.90.0/26 is directly connected, FastEthernet0/0
L       200.200.90.62/32 is directly connected, FastEthernet0/0
S       200.200.90.64/27 [1/0] via 200.200.90.130
C       200.200.90.128/30 is directly connected, FastEthernet0/1
L       200.200.90.129/32 is directly connected, FastEthernet0/1
S       200.200.90.132/30 [1/0] via 200.200.90.130

RT-Taylor-PS>
```

Topologia 2 — topo2-rotaTaylor.png

Topologia 3

RT-Taylor – Tabela de Roteamento				
Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.0	255.255.255.192 (/26)	—	Fa0/0
C	200.200.90.128	255.255.255.252 (/30)	—	Fa0/1
S	200.200.90.64	255.255.255.224 (/27)	200.200.90.130	Fa0/1
S	200.200.90.96	255.255.255.224 (/27)	200.200.90.130	Fa0/1

RT-Postel – Tabela de Roteamento				
Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.64	255.255.255.224 (/27)	—	Fa0/1
C	200.200.90.96	255.255.255.224 (/27)	—	Fa1/0
C	200.200.90.128	255.255.255.252 (/30)	—	Fa0/0
S	200.200.90.0	255.255.255.192 (/26)	200.200.90.129	Fa0/0

Comprovação no Packet Tracer (show ip route)


```
RT-Postel-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 7 subnets, 4 masks
S       200.200.90.0/26 [1/0] via 200.200.90.129
C       200.200.90.64/27 is directly connected, FastEthernet0/1
L       200.200.90.94/32 is directly connected, FastEthernet0/1
C       200.200.90.96/27 is directly connected, FastEthernet1/0
L       200.200.90.126/32 is directly connected, FastEthernet1/0
C       200.200.90.128/30 is directly connected, FastEthernet0/0
L       200.200.90.130/32 is directly connected, FastEthernet0/0

RT-Postel-PS>
```

Topologia 3 — topo3-rotaPostel.png

```
RT-Taylor-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 6 subnets, 4 masks
C       200.200.90.0/26 is directly connected, FastEthernet0/0
L       200.200.90.62/32 is directly connected, FastEthernet0/0
S       200.200.90.64/27 [1/0] via 200.200.90.130
S       200.200.90.96/27 [1/0] via 200.200.90.130
C       200.200.90.128/30 is directly connected, FastEthernet0/1
L       200.200.90.129/32 is directly connected, FastEthernet0/1

RT-Taylor-PS>
```

Topologia 3 — topo3-rotaTaylor.png

Topologia 4

RT-Taylor – Tabela de Roteamento

Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.0	255.255.255.192 (/26)	—	Fa0/0
C	200.200.90.128	255.255.255.252 (/30)	—	Fa0/1
C	200.200.90.136	255.255.255.252 (/30)	—	Fa1/0
S	200.200.90.64	255.255.255.224 (/27)	200.200.90.130	Fa0/1
S	200.200.90.96	255.255.255.224 (/27)	200.200.90.138	Fa1/0
S	200.200.90.132	255.255.255.252 (/30)	200.200.90.130	Fa0/1

RT-Postel – Tabela de Roteamento

Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.64	255.255.255.224 (/27)	—	Fa0/1
C	200.200.90.128	255.255.255.252 (/30)	—	Fa0/0
C	200.200.90.132	255.255.255.252 (/30)	—	Fa1/0
S	200.200.90.0	255.255.255.192 (/26)	200.200.90.129	Fa0/0
S	200.200.90.96	255.255.255.224 (/27)	200.200.90.134	Fa1/0

RT-Postel – Tabela de Roteamento

Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
S	200.200.90.136	255.255.255.252 (/30)	200.200.90.129	Fa0/0

RT-Babbage – Tabela de Roteamento

Tipo	Rede de Destino	Máscara	Next Hop	Interface de Saída
C	200.200.90.96	255.255.255.224 (/27)	—	Fa1/0
C	200.200.90.132	255.255.255.252 (/30)	—	Fa0/0
C	200.200.90.136	255.255.255.252 (/30)	—	Fa0/1
S	200.200.90.0	255.255.255.192 (/26)	200.200.90.137	Fa0/1
S	200.200.90.64	255.255.255.224 (/27)	200.200.90.133	Fa0/0
S	200.200.90.128	255.255.255.252 (/30)	200.200.90.137	Fa0/1

Comprovação no Packet Tracer (show ip route)

```
RT-Babbage-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 9 subnets, 4 masks
S       200.200.90.0/26 [1/0] via 200.200.90.137
S       200.200.90.64/27 [1/0] via 200.200.90.133
C       200.200.90.96/27 is directly connected, FastEthernet1/0
L       200.200.90.126/32 is directly connected, FastEthernet1/0
S       200.200.90.128/30 [1/0] via 200.200.90.137
C       200.200.90.132/30 is directly connected, FastEthernet0/0
L       200.200.90.134/32 is directly connected, FastEthernet0/0
C       200.200.90.136/30 is directly connected, FastEthernet0/1
L       200.200.90.138/32 is directly connected, FastEthernet0/1

RT-Babbage-PS>
```

Topologia 4 — topo4-rotaBabbage.png

```
RT-Postel-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 9 subnets, 4 masks
S       200.200.90.0/26 [1/0] via 200.200.90.129
C       200.200.90.64/27 is directly connected, FastEthernet0/1
L       200.200.90.94/32 is directly connected, FastEthernet0/1
S       200.200.90.96/27 [1/0] via 200.200.90.134
C       200.200.90.128/30 is directly connected, FastEthernet0/0
L       200.200.90.130/32 is directly connected, FastEthernet0/0
C       200.200.90.132/30 is directly connected, FastEthernet1/0
L       200.200.90.133/32 is directly connected, FastEthernet1/0
S       200.200.90.136/30 [1/0] via 200.200.90.129

RT-Postel-PS>
```

Topologia 4 — topo4-rotaPostel.png

```
RT-Taylor-PS>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    200.200.90.0/24 is variably subnetted, 9 subnets, 4 masks
C       200.200.90.0/26 is directly connected, FastEthernet0/0
L       200.200.90.62/32 is directly connected, FastEthernet0/0
S       200.200.90.64/27 [1/0] via 200.200.90.130
S       200.200.90.96/27 [1/0] via 200.200.90.138
C       200.200.90.128/30 is directly connected, FastEthernet0/1
L       200.200.90.129/32 is directly connected, FastEthernet0/1
S       200.200.90.132/30 [1/0] via 200.200.90.130
C       200.200.90.136/30 is directly connected, FastEthernet1/0
L       200.200.90.137/32 is directly connected, FastEthernet1/0

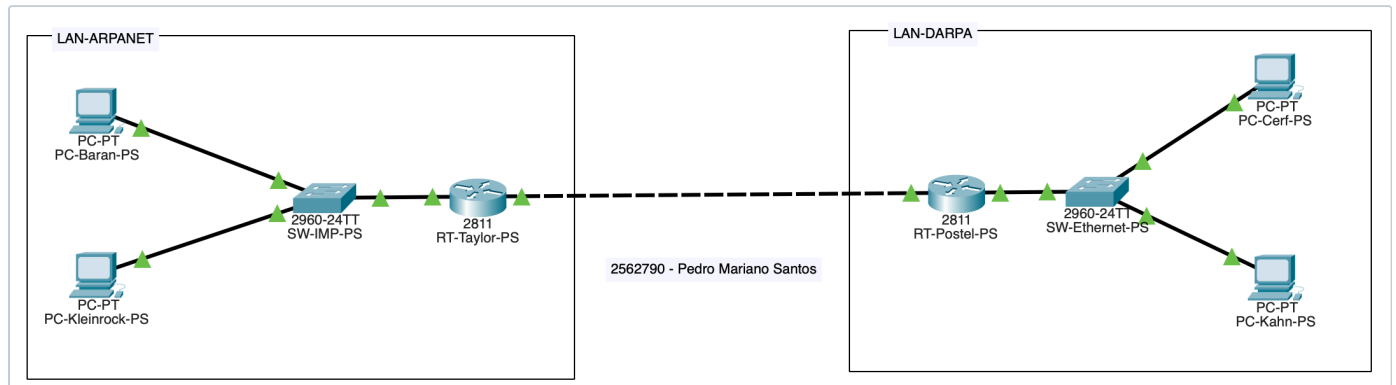
RT-Taylor-PS>
```

Topologia 4 — topo4-rotaTaylor.png

4. Tarefa 3 – Topologias no Packet Tracer

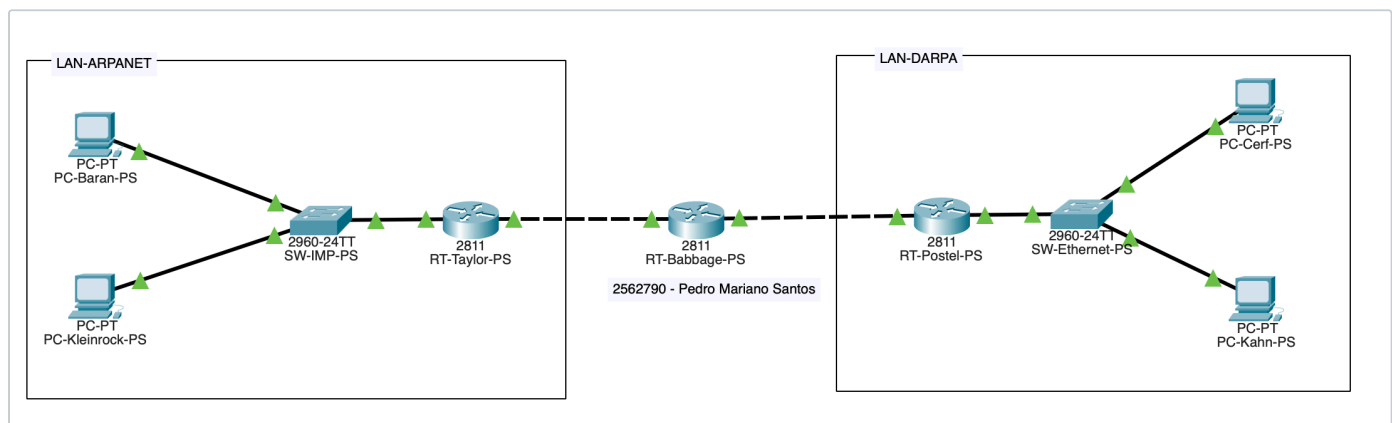
Topologias funcionais montadas no Cisco Packet Tracer (roteadores 2811, switches 2960), com os dispositivos nomeados (sufixo **-PS**) e a identificação do aluno na área de trabalho.

Topologia 1 – Duas LANs interligadas por dois roteadores



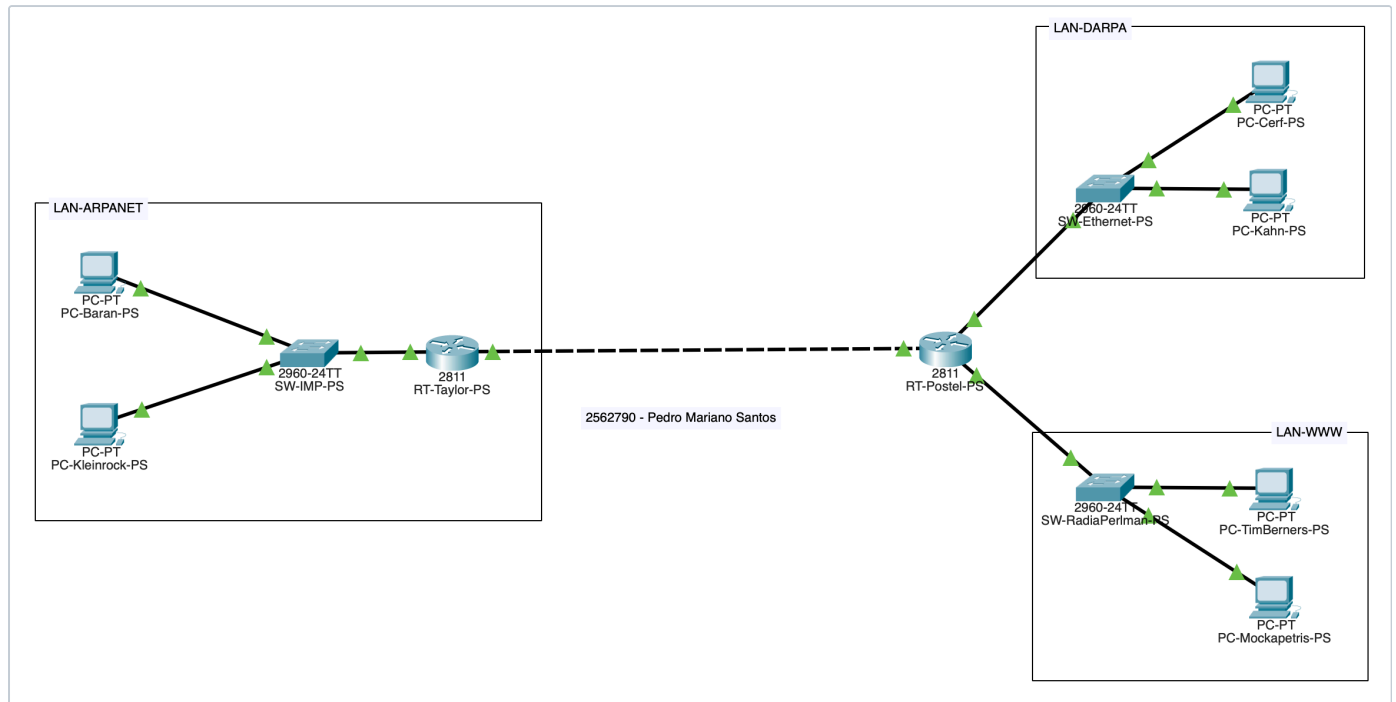
Topologia 1 — LAN-ARPANET e LAN-DARPA via RT-Taylor e RT-Postel (WAN-IMP-NET)

Topologia 2 – Roteador intermediário (RT-Babbage)



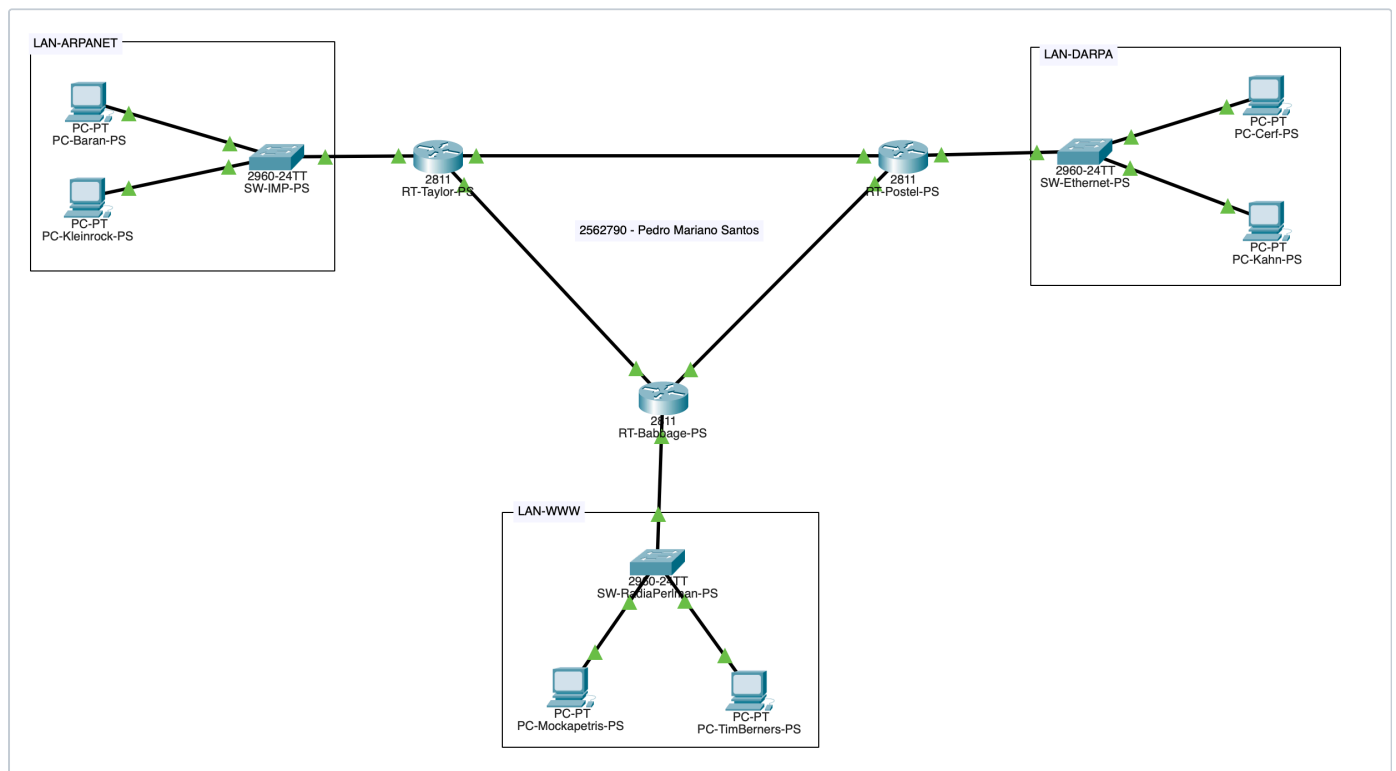
Topologia 2 — RT-Taylor ↔ RT-Babbage ↔ RT-Postel (WAN-IMP-NET e WAN-MILNET)

Topologia 3 – Terceira LAN (LAN-WWW) no RT-Postel



Topologia 3 — LAN-WWW adicionada ao RT-Postel via módulo NM-1FE2W (Fa1/0)

Topologia 4 – Topologia em triângulo



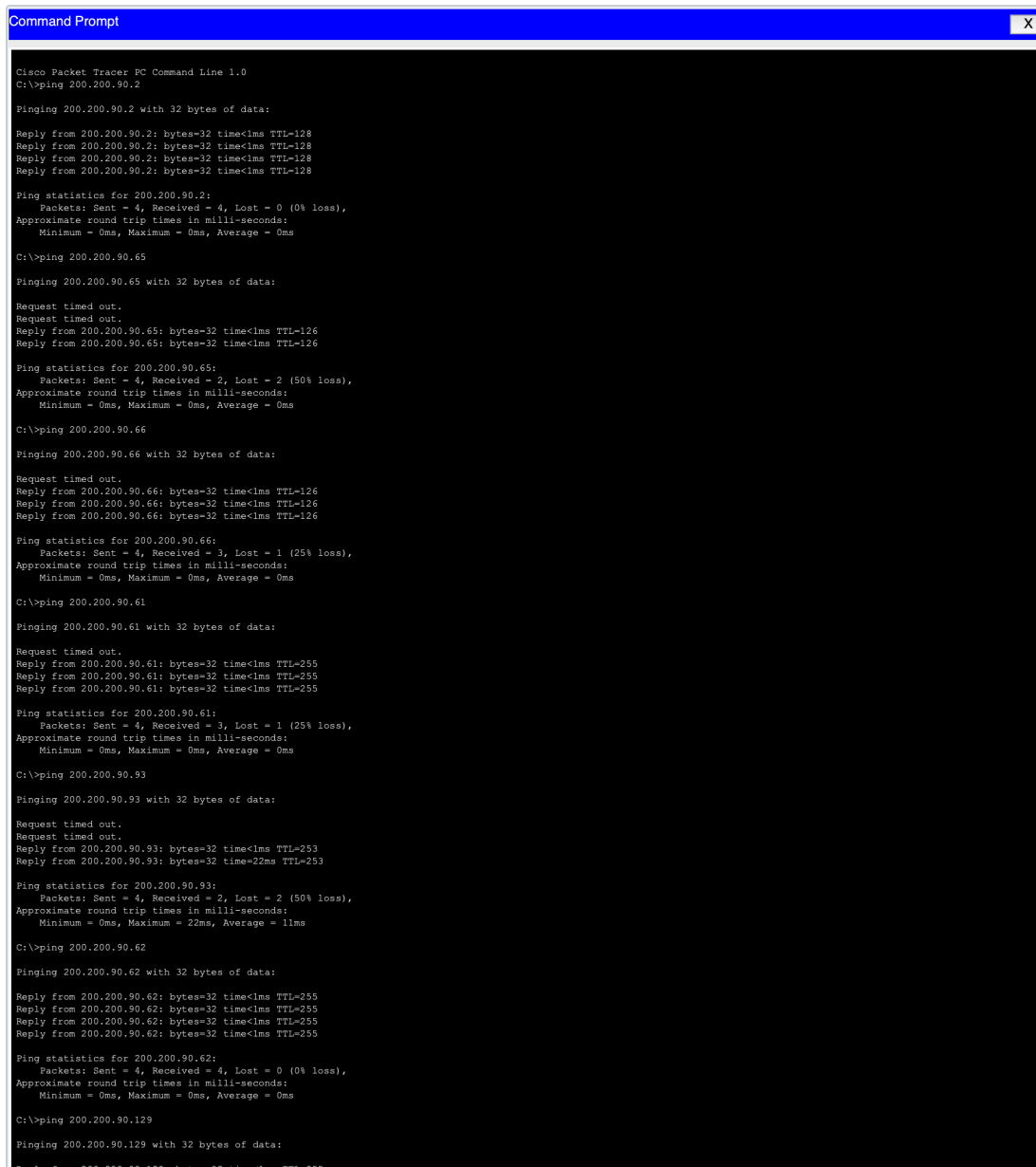
Topologia 4 — RT-Taylor, RT-Postel e RT-Babbage interligados em triângulo; LAN-WWW no RT-Babbage

5. Checklist de Testes de Conectividade (Apêndice 3)

Testes de ping IPv4 executados a partir do **PC-Baran** para todas as interfaces dos dispositivos finais e intermediários de cada topologia. As tabelas indicam o resultado e as capturas de tela comprovam o status de sucesso.

Topologia 1

Dispositivo	Interface	IPv4	Resultado do Ping (origem PC-Baran)
PC-Baran	NIC	200.200.90.1	Origem do teste
PC-Kleinrock	NIC	200.200.90.2	Sucesso ✓
PC-Cerf	NIC	200.200.90.65	Sucesso ✓
PC-Kahn	NIC	200.200.90.66	Sucesso ✓
SW-IMP	SVI	200.200.90.61	Sucesso ✓
SW-Ethernet	SVI	200.200.90.93	Sucesso ✓
RT-Taylor	Fa0/0	200.200.90.62	Sucesso ✓
RT-Taylor	Fa0/1	200.200.90.129	Sucesso ✓
RT-Postel	Fa0/0	200.200.90.130	Sucesso ✓
RT-Postel	Fa0/1	200.200.90.94	Sucesso ✓



```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.200.90.2

Pinging 200.200.90.2 with 32 bytes of data:

Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128

Ping statistics for 200.200.90.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.65

Pinging 200.200.90.65 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 200.200.90.65: bytes=32 time<1ms TTL=126
Reply from 200.200.90.65: bytes=32 time<1ms TTL=126

Ping statistics for 200.200.90.65:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.66

Pinging 200.200.90.66 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.66: bytes=32 time<1ms TTL=126
Reply from 200.200.90.66: bytes=32 time<1ms TTL=126
Reply from 200.200.90.66: bytes=32 time<1ms TTL=126

Ping statistics for 200.200.90.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.61

Pinging 200.200.90.61 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.61:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.93

Pinging 200.200.90.93 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 200.200.90.93: bytes=32 time<1ms TTL=253
Reply from 200.200.90.93: bytes=32 time=22ms TTL=253

Ping statistics for 200.200.90.93:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 22ms, Average = 11ms

C:\>ping 200.200.90.62

Pinging 200.200.90.62 with 32 bytes of data:

Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.129

Pinging 200.200.90.129 with 32 bytes of data:

Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
```

Topologia 1 — testes de ping a partir do PC-Baran — topo1-pings1:2.png

```
C:\>ping 200.200.90.129

Pinging 200.200.90.129 with 32 bytes of data:

Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 19ms, Average = 8ms

C:\>ping 200.200.90.130

Pinging 200.200.90.130 with 32 bytes of data:

Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.94

Pinging 200.200.90.94 with 32 bytes of data:

Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time=23ms TTL=254

Ping statistics for 200.200.90.94:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 23ms, Average = 5ms

C:\>
```

Topologia 1 — testes de ping a partir do PC-Baran — topo1-pings2:2.png

Topologia 2

Dispositivo	Interface	IPv4	Resultado do Ping (origem PC-Baran)
PC-Baran	NIC	200.200.90.1	Origem do teste
PC-Kleinrock	NIC	200.200.90.2	Sucesso ✓
PC-Cerf	NIC	200.200.90.65	Sucesso ✓
PC-Kahn	NIC	200.200.90.66	Sucesso ✓
SW-IMP	SVI	200.200.90.61	Sucesso ✓
SW-Ethernet	SVI	200.200.90.93	Sucesso ✓
RT-Taylor	Fa0/0	200.200.90.62	Sucesso ✓
RT-Taylor	Fa0/1	200.200.90.129	Sucesso ✓
RT-Postel	Fa0/0	200.200.90.134	Sucesso ✓
RT-Postel	Fa0/1	200.200.90.94	Sucesso ✓
RT-Babbage	Fa0/0	200.200.90.130	Sucesso ✓
RT-Babbage	Fa0/1	200.200.90.133	Sucesso ✓


```
Pinging 200.200.90.129 with 32 bytes of data:

Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.130

Pinging 200.200.90.130 with 32 bytes of data:

Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time=22ms TTL=254

Ping statistics for 200.200.90.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 22ms, Average = 5ms

C:\>ping 200.200.90.133

Pinging 200.200.90.133 with 32 bytes of data:

Reply from 200.200.90.133: bytes=32 time<1ms TTL=254
Reply from 200.200.90.133: bytes=32 time<1ms TTL=254
Reply from 200.200.90.133: bytes=32 time<1ms TTL=254
Reply from 200.200.90.133: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.133:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.134

Pinging 200.200.90.134 with 32 bytes of data:

Reply from 200.200.90.134: bytes=32 time<1ms TTL=253
Reply from 200.200.90.134: bytes=32 time<1ms TTL=253
Reply from 200.200.90.134: bytes=32 time<1ms TTL=253
Reply from 200.200.90.134: bytes=32 time<1ms TTL=253

Ping statistics for 200.200.90.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.94

Pinging 200.200.90.94 with 32 bytes of data:

Reply from 200.200.90.94: bytes=32 time<1ms TTL=253
Reply from 200.200.90.94: bytes=32 time<1ms TTL=253
Reply from 200.200.90.94: bytes=32 time=1ms TTL=253
Reply from 200.200.90.94: bytes=32 time=17ms TTL=253

Ping statistics for 200.200.90.94:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms

C:\>
```

Topologia 2 — testes de ping a partir do PC-Baran — topo2-ping2:2.png

```
Command Prompt
C:\>
C:\>ping 200.200.90.2

Pinging 200.200.90.2 with 32 bytes of data:

Reply from 200.200.90.2: bytes=32 time=19ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128

Ping statistics for 200.200.90.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 19ms, Average = 4ms

C:\>ping 200.200.90.65

Pinging 200.200.90.65 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 200.200.90.65: bytes=32 time<1ms TTL=125
Reply from 200.200.90.65: bytes=32 time<1ms TTL=125

Ping statistics for 200.200.90.65:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
C:\>ping 200.200.90.66

Pinging 200.200.90.66 with 32 bytes of data:

Reply from 200.200.90.66: bytes=32 time=18ms TTL=125
Reply from 200.200.90.66: bytes=32 time<1ms TTL=125
Reply from 200.200.90.66: bytes=32 time<1ms TTL=125
Reply from 200.200.90.66: bytes=32 time<1ms TTL=125

Ping statistics for 200.200.90.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 4ms

C:\>ping 200.200.90.61

Pinging 200.200.90.61 with 32 bytes of data:

Reply from 200.200.90.61: bytes=32 time<1ms TTL=255
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.61:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.93

Pinging 200.200.90.93 with 32 bytes of data:

Reply from 200.200.90.93: bytes=32 time<1ms TTL=252
Reply from 200.200.90.93: bytes=32 time<1ms TTL=252
Reply from 200.200.90.93: bytes=32 time<1ms TTL=252
Reply from 200.200.90.93: bytes=32 time<1ms TTL=252

Ping statistics for 200.200.90.93:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.62

Pinging 200.200.90.62 with 32 bytes of data:

Reply from 200.200.90.62: bytes=32 time=18ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time=17ms TTL=255

Ping statistics for 200.200.90.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 8ms

C:\>ping 200.200.90.129

Pinging 200.200.90.129 with 32 bytes of data:

Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
```

Topologia 2 — testes de ping a partir do PC-Baran — topo2-pings1:2.png

Topologia 3

Dispositivo	Interface	IPv4	Resultado do Ping (origem PC-Baran)
PC-Baran	NIC	200.200.90.1	Origem do teste
PC-Kleinrock	NIC	200.200.90.2	Sucesso ✓
PC-Cerf	NIC	200.200.90.65	Sucesso ✓
PC-Kahn	NIC	200.200.90.66	Sucesso ✓
PC-TimBerners	NIC	200.200.90.97	Sucesso ✓
PC-Mockapetris	NIC	200.200.90.98	Sucesso ✓
SW-IMP	SVI	200.200.90.61	Sucesso ✓
SW-Ethernet	SVI	200.200.90.93	Sucesso ✓
SW-RadiaPerlman	SVI	200.200.90.125	Sucesso ✓

Dispositivo	Interface	IPv4	Resultado do Ping (origem PC-Baran)
RT-Taylor	Fa0/0	200.200.90.62	Sucesso ✓
RT-Taylor	Fa0/1	200.200.90.129	Sucesso ✓
RT-Postel	Fa0/0	200.200.90.130	Sucesso ✓
RT-Postel	Fa0/1	200.200.90.94	Sucesso ✓
RT-Postel	Fa1/0	200.200.90.126	Sucesso ✓

```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 200.200.90.2

Pinging 200.200.90.2 with 32 bytes of data:

Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128

Ping statistics for 200.200.90.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.65

Pinging 200.200.90.65 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 200.200.90.65: bytes=32 time<1ms TTL=126
Reply from 200.200.90.65: bytes=32 time<1ms TTL=126

Ping statistics for 200.200.90.65:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.66

Pinging 200.200.90.66 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.66: bytes=32 time<1ms TTL=126
Reply from 200.200.90.66: bytes=32 time<1ms TTL=126
Reply from 200.200.90.66: bytes=32 time<1ms TTL=126

Ping statistics for 200.200.90.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.97

Pinging 200.200.90.97 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.97: bytes=32 time<1ms TTL=126
Reply from 200.200.90.97: bytes=32 time<1ms TTL=126
Reply from 200.200.90.97: bytes=32 time<1ms TTL=126

Ping statistics for 200.200.90.97:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 200.200.90.98

Pinging 200.200.90.98 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.98: bytes=32 time<1ms TTL=126
Reply from 200.200.90.98: bytes=32 time<1ms TTL=126
Reply from 200.200.90.98: bytes=32 time<1ms TTL=126

Ping statistics for 200.200.90.98:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.93

Pinging 200.200.90.93 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 200.200.90.93: bytes=32 time=16ms TTL=253
Reply from 200.200.90.93: bytes=32 time<1ms TTL=253

Ping statistics for 200.200.90.93:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 8ms

C:\>ping 200.200.90.125

Pinging 200.200.90.125 with 32 bytes of data:
```

Topologia 3 — testes de ping a partir do PC-Baran — topo3-pings1:2.png

```
Pinging 200.200.90.125 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 200.200.90.125: bytes=32 time<1ms TTL=253
Reply from 200.200.90.125: bytes=32 time<1ms TTL=253

Ping statistics for 200.200.90.125:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.62

Pinging 200.200.90.62 with 32 bytes of data:

Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.129

Pinging 200.200.90.129 with 32 bytes of data:

Reply from 200.200.90.129: bytes=32 time=20ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255
Reply from 200.200.90.129: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>ping 200.200.90.130

Pinging 200.200.90.130 with 32 bytes of data:

Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time=20ms TTL=254

Ping statistics for 200.200.90.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>ping 200.200.90.94

Pinging 200.200.90.94 with 32 bytes of data:

Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.94:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.126

Pinging 200.200.90.126 with 32 bytes of data:

Reply from 200.200.90.126: bytes=32 time<1ms TTL=254
Reply from 200.200.90.126: bytes=32 time<1ms TTL=254
Reply from 200.200.90.126: bytes=32 time<1ms TTL=254
Reply from 200.200.90.126: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Topologia 3 — testes de ping a partir do PC-Baran — topo3-pings2:2.png

Topologia 4

Dispositivo	Interface	IPv4	Resultado do Ping (origem PC-Baran)
PC-Baran	NIC	200.200.90.1	Origem do teste
PC-Kleinrock	NIC	200.200.90.2	Sucesso ✓
PC-Cerf	NIC	200.200.90.65	Sucesso ✓
PC-Kahn	NIC	200.200.90.66	Sucesso ✓
PC-TimBerners	NIC	200.200.90.97	Sucesso ✓
PC-Mockapetris	NIC	200.200.90.98	Sucesso ✓
SW-IMP	SVI	200.200.90.61	Sucesso ✓
SW-Ethernet	SVI	200.200.90.93	Sucesso ✓
SW-RadiaPerlman	SVI	200.200.90.125	Sucesso ✓
RT-Taylor	Fa0/0	200.200.90.62	Sucesso ✓
RT-Taylor	Fa0/1	200.200.90.129	Sucesso ✓
RT-Taylor	Fa1/0	200.200.90.137	Sucesso ✓

Dispositivo	Interface	IPv4	Resultado do Ping (origem PC-Baran)
RT-Postel	Fa0/0	200.200.90.130	Sucesso ✓
RT-Postel	Fa0/1	200.200.90.94	Sucesso ✓
RT-Postel	Fa1/0	200.200.90.133	Sucesso ✓
RT-Babbage	Fa0/0	200.200.90.134	Sucesso ✓
RT-Babbage	Fa0/1	200.200.90.138	Sucesso ✓
RT-Babbage	Fa1/0	200.200.90.126	Sucesso ✓

```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.200.90.2

Pinging 200.200.90.2 with 32 bytes of data:

Reply from 200.200.90.2: bytes=32 time=21ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128
Reply from 200.200.90.2: bytes=32 time<1ms TTL=128

Ping statistics for 200.200.90.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 5ms

C:\>ping 200.200.90.65

Pinging 200.200.90.65 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 200.200.90.65: bytes=32 time=19ms TTL=126
Reply from 200.200.90.65: bytes=32 time=10ms TTL=126

Ping statistics for 200.200.90.65:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 19ms, Average = 14ms

C:\>ping 200.200.90.66

Pinging 200.200.90.66 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.66: bytes=32 time<1ms TTL=126
Reply from 200.200.90.66: bytes=32 time<1ms TTL=126
Reply from 200.200.90.66: bytes=32 time=9ms TTL=126

Ping statistics for 200.200.90.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 3ms
```

Topologia 4 — testes de ping a partir do PC-Baran — topo4-pings0.png

```
Command Prompt X

Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 200.200.90.97

Pinging 200.200.90.97 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 200.200.90.97: bytes=32 time=14ms TTL=126
Reply from 200.200.90.97: bytes=32 time=32ms TTL=126

Ping statistics for 200.200.90.97:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 14ms, Maximum = 32ms, Average = 23ms

C:\>ping 200.200.90.98

Pinging 200.200.90.98 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.98: bytes=32 time<1ms TTL=126
Reply from 200.200.90.98: bytes=32 time<1ms TTL=126
Reply from 200.200.90.98: bytes=32 time=20ms TTL=126

Ping statistics for 200.200.90.98:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 0ms, Maximum = 20ms, Average = 6ms

C:\>ping 200.200.90.61

Pinging 200.200.90.61 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255
Reply from 200.200.90.61: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.61:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.93

Pinging 200.200.90.93 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.93: bytes=32 time<1ms TTL=253
Reply from 200.200.90.93: bytes=32 time<1ms TTL=253
Reply from 200.200.90.93: bytes=32 time<1ms TTL=253

Ping statistics for 200.200.90.93:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.125

Pinging 200.200.90.125 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.125: bytes=32 time<1ms TTL=253
Reply from 200.200.90.125: bytes=32 time<1ms TTL=253
Reply from 200.200.90.125: bytes=32 time=19ms TTL=253

Ping statistics for 200.200.90.125:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 0ms, Maximum = 19ms, Average = 6ms

C:\>ping 200.200.90.62

Pinging 200.200.90.62 with 32 bytes of data:

Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255
Reply from 200.200.90.62: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
      Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.129

Pinging 200.200.90.129 with 32 bytes of data:
```

Topologia 4 — testes de ping a partir do PC-Baran — topo4-pings1:2.png

```
Command Prompt X

Reply from 200.200.90.137: bytes=32 time<1ms TTL=255

Ping statistics for 200.200.90.137:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.130

Pinging 200.200.90.130 with 32 bytes of data:

Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254
Reply from 200.200.90.130: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.200.90.94

Pinging 200.200.90.94 with 32 bytes of data:

Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time<1ms TTL=254
Reply from 200.200.90.94: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.94:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 200.200.90.133

Pinging 200.200.90.133 with 32 bytes of data:

Reply from 200.200.90.133: bytes=32 time<1ms TTL=254
Reply from 200.200.90.133: bytes=32 time<1ms TTL=254
Reply from 200.200.90.133: bytes=32 time<1ms TTL=254
Reply from 200.200.90.133: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.133:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 200.200.90.134

Pinging 200.200.90.134 with 32 bytes of data:

Request timed out.
Reply from 200.200.90.134: bytes=32 time=2ms TTL=254
Reply from 200.200.90.134: bytes=32 time=2ms TTL=254
Reply from 200.200.90.134: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.134:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 200.200.90.138

Pinging 200.200.90.138 with 32 bytes of data:

Reply from 200.200.90.138: bytes=32 time<1ms TTL=254
Reply from 200.200.90.138: bytes=32 time<1ms TTL=254
Reply from 200.200.90.138: bytes=32 time<1ms TTL=254
Reply from 200.200.90.138: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.138:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 200.200.90.126

Pinging 200.200.90.126 with 32 bytes of data:

Reply from 200.200.90.126: bytes=32 time=13ms TTL=254
Reply from 200.200.90.126: bytes=32 time=17ms TTL=254
Reply from 200.200.90.126: bytes=32 time<1ms TTL=254
Reply from 200.200.90.126: bytes=32 time<1ms TTL=254

Ping statistics for 200.200.90.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 7ms
```

Topologia 4 — testes de ping a partir do PC-Baran — topo4-pings2:2.png